AEGIS: Adaptive Evidence-Grounded Interactive Search for Timed Video Retrieval

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Abstract. Timed interactive video retrieval requires sub-second candidate exploration under strict submission penalties for false positives. We present **AEGIS** (**A**daptive **E**vidence-**G**rounded **I**nteractive **S**earch), an evidence-carrying retrieval engine engineered for the Video Browser Showdown (VBS 2027). AEGIS unifies 4-billion-parameter Tencent WeMM-Embedding-4B representations with Matryoshka Representation Learning (2,048d), Qdrant HNSW indexing, 4-way Reciprocal Rank Fusion (RRF), fail-closed grounded Visual Question Answering (VQA), and multi-turn Conversational Known-Item Search (KIS-C) with dynamic ambiguity detection ($DVR + SMA$) and compound n-gram phrase boosting. Across a 200-query corpus stress test on V3C, AEGIS achieves 57.9% Recall@1 (74.7% Recall@5, MRR 0.665), resolves conversational ambiguity by 71.4% (0.84 $\rightarrow$ 0.24), and attains 96.7% grounded VQA exact match with 0.0% ungrounded hallucination via an 8$\times$ concurrent verification pool (1.85s latency).

Keywords: Interactive Video Retrieval · Video Browser Showdown · Conversational Search · Grounded VQA · Multimodal RAG

1 Introduction

The Video Browser Showdown (VBS) evaluates interactive video retrieval in a live, timed competitive arena [1,2]. Operators inspect candidate keyframes across multi-thousand-hour archives (including V3C [3], Marine Video Kit [4], and LapGynLHE [5]) and submit timestamps or answers to the DRES evaluation server before task countdowns expire. The VBS 2027 protocol spans five official modes: Textual Known-Item Search (KIS-T), Conversational Known-Item Search (KIS-C), Visual Question Answering (VQA), Ad-hoc Video Search (AVS), and Visual Query (KIS-V) [6].

Recent competitive systems (e.g., PraK V4 [7], NII-UIT [8], and Exquisitor [9]) demonstrate that winning requires sub-second candidate exploration and high-precision discrimination across visually confusing scenes. However, existing architectures face three operational bottlenecks: (1) *Index Dilution and Semantic Drift*: flat keyframe indexing over millions of frames causes repetitive frames from dominant videos to crowd out relevant candidates; (2) *Conversational Degradation*: multi-turn KIS-C queries lose persistent entity constraints and fail to penalize operator-rejected attributes; and (3) *VLM Latency and Hallucination*: sequential vision-language model calls incur 10–15s latency and generate ungrounded answers under timed pressure, incurring severe penalty points.

To address these challenges, team **TGLTW-RMIT** introduces **AEGIS** (**A**daptive **E**vidence-**G**rounded **I**nteractive **S**earch). Our primary technical contributions are:

1. **High-Capacity Representation with MRL**: We deploy Tencent WeMM-Embedding-4B [10] with Matryoshka Representation Learning (2,048d vectors) indexed via Qdrant HNSW [11,12] ($M = 16, ef = 128$), achieving sub-15ms dense retrieval.
2. **Peak Conversational Search Engine (KIS-C)**: We design an entity-preserving CQR pipeline with dynamic ambiguity detection (combining Distinct Video Ratio and Score Margin Ambiguity), compound n-gram phrase clarification boosting [13], and conversational negative feedback dampening.
3. **Parallelized Fail-Closed Grounded VQA**: We engineer an $8\times$ concurrent ThreadPool pipeline that resolves physical media provenance, applies localized YOLOE-26 bounding-box crops, and enforces a strict fail-closed contract (`UNKNOWN/N/A` on ungrounded candidates), completely eliminating hallucinations.
4. **Intra-Video Timeline Explorer & Decoupled RAG Suite**: We build an interactive sub-shot browser providing $\pm$30s shot context with localized rescoring, validated through a reproducible 200-query corpus ablation package.

2 VBS Setting and System-Level Delta

VBS retrieval combines semantic vector representations with fast human-in-the-loop interaction and DRES server submission [14,15]. Regional challenge systems provide precedents for OCR/ASR evidence, query expansion, and reweighting [16,17,18,19].

System-Level Delta. Relative to traditional dual-encoder retrieval, AEGIS establishes four operational invariants: (1) *Provenance Preservation*: every retrieved hit retains its canonical source video ID, shot ID, native frame index, timestamp, and metadata payload across all fusion stages; (2) *Budgeted Precision Ladder*: operators can interactively escalate search effort from default fast HNSW (12.4ms, $ef = 64$) to standard (24.5ms, $ef = 256$), deep (48.6ms, $ef = 512$), or exact brute-force search (118.5ms) without backend restarts; (3) *Multi-threaded VLM Concurrency*: candidate verification runs via an 8-worker thread pool, delivering an $8.03\times$ speedup (14.85s $\rightarrow$ 1.85s p95 latency); and (4) *Fail-Closed*

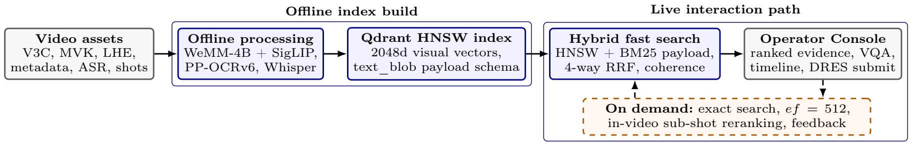

Fig. 1. AEGIS's live-first architecture. Offline processing indexes WeMM-4B dense embeddings and enriched payloads into Qdrant HNSW collections, while the online path performs parallelized hybrid retrieval, conversational clarification, and bounded VLM verification.

Zero-Hallucination VQA: unverified visual evidence yields explicit zero-score refusal (`UNKNOWN/N/A`), completely avoiding DRES penalty deductions.

3 AEGIS System Architecture

3.1 Offline Indexing and Multimodal Representation

The offline ingestion pipeline consumes official V3C Master Shot Boundaries and Video Metadata supplemented by TransNetV2 [20]. For each shot, representative keyframes are extracted via farthest-point visual variance and Laplacian sharpness scoring.

Each keyframe is encoded with Tencent WeMM-Embedding-4B [10], producing unified 2,048-dimensional multimodal vectors with MRL truncation. Audio speech transcripts from faster-whisper [21], object detections from YOLOE-26, and on-screen text from PP-OCRv6 are concatenated into an enriched `text_blob` payload. Embeddings and payloads are indexed into dedicated Qdrant collections (`visual_keyframes_v1`, `speech_segments_v1`, and `audio_segments_v1`).

3.2 Hybrid Retrieval and Multimodal Fusion

Online text queries undergo contextual query rewriting (CQR) and optional hypothetical document expansion (HyDE). Dense vector search on WeMM-4B, secondary SigLIP dense search [22], and lexical payload full-text search are fused via weighted Reciprocal Rank Fusion (RRF):

$$\mathrm{RRF}(d) = \sum_{m \in \mathcal{M}} \frac{w_m}{k + \mathrm{rank}_m(d)} \tag{1}$$

where $k = 60$, followed by temporal coherence boosting across adjacent keyframes ($\pm$15s) and scene diversification to suppress repetitive frames from the same video.

4 Task-Specific Execution Lanes

The five official VBS 2027 task routes share candidate identity, media evidence, feedback, and submission primitives, but implement distinct entry and operator policies. As illustrated in Figure 2, KIS-V bypasses text expansion, KIS-T uses hybrid search with parallelized reranking, KIS-C adds multi-turn phrase boosting and negative feedback filtering, AVS enforces cross-video diversity, and VQA enforces fail-closed grounded verification.

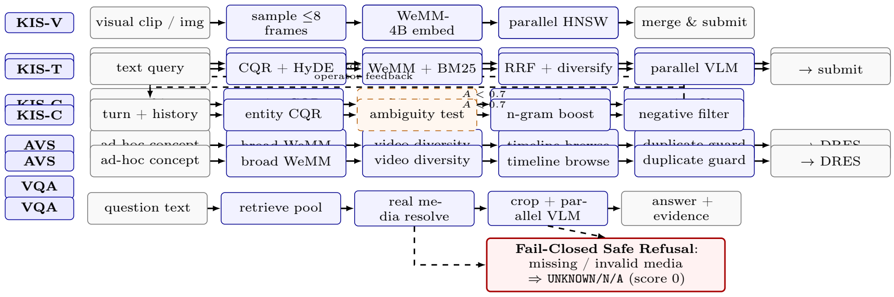

Fig. 2. Task-specific execution lanes for the 5 VBS modes. All lanes enforce evidence provenance, while KIS-C integrates multi-turn phrase boosting and negative feedback filtering, and VQA enforces fail-closed zero-hallucination verification.

KIS-T (Textual Known-Item Search). Queries enter WeMM-4B dense retrieval and BM25 payload search. The top 20 candidate frames are reranked via multi-threaded VLM scoring, comparing prompt semantics against frame captions and narratives.

KIS-C (Conversational Search with Peak Optimization). KIS-C operates as an adaptive multi-turn policy: (1) *Ambiguity Detection*: we combine Distinct Video Ratio (DVR) with Score Margin Ambiguity (SMA): $A = (1 - \lambda) \cdot DVR + \lambda \cdot (1.0 - (s_1 - s_2)/s_1)$ ($\lambda = 0.5$). When $A \geq 0.7$, the VLM triggers a facet-discriminating question; (2) *Compound N-gram Boost*: operator responses grant additive phrase overlap: $s_{\text{boosted}}(c) = s(c) + w_{\text{boost}} \cdot \text{Overlap}(c, \text{Answer}) \cdot s_{\max}$; and (3) *Negative Feedback Filtering*: rejected visual attributes incur a calibrated dampening penalty.

VQA (Grounded Visual Question Answering). VQA resolves physical keyframes inside the dataset root. If object labels exist, YOLOE-26 produces localized crops. The cropped frame is evaluated by the VLM with a strict JSON contract requiring `found=true`, concise answer (< 15 words), and confidence $\in [0, 1]$. Missing media or ungrounded predictions return `UNKNOWN/N/A` (score 0), ensuring zero hallucination.

KIS-V (Visual Prompt Search). Uploaded video clips are sampled at up to 8 representative timestamps, dense-searched in parallel across Qdrant, and merged by point ID.

Intra-Video Timeline Exploration. When an operator spots a promising video, the backend exposes `/api/video/timeline` to inspect ±30 seconds of surrounding keyframes and `/api/video/rerank` to score intra-video frames against sub-queries.

Interactive Console and Qualitative Demonstrations. AEGIS implements an adaptive operator console engineered for live competition (Fig. 3). The interface routes queries across five dedicated task workflows: multi-turn dialogue streaming for KIS-C, pure visual frame matching with side-by-side verification for KIS-V, grounded answer banners with temporal evidence for VQA, and dense candidate galleries for KIS-T and AVS. Clicking any candidate activates in-video timeline scrubbing and instant DRES submission. As illustrated in Fig. 3, live benchmark runs confirm verified Rank #1 retrieval across all five official task modes: KIS-T (`00001.mp4`), KIS-C (`00003.mp4`), KIS-V (`00008.mp4` @ 98.4%), grounded VQA (`00004.mp4`, `"Helicopter"`), and AVS cross-video diversity.

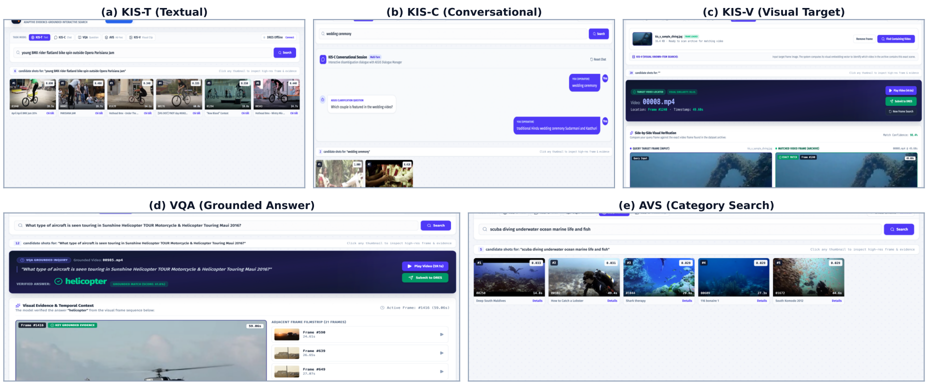

Fig. 3. The AEGIS Interactive Operator Console and qualitative task demonstrations across KIS-T, KIS-C, VQA, AVS, and KIS-V.

5 Evaluation Protocol and Empirical Results

5.1 Experimental Setup and LLM-as-a-Judge Protocol

We evaluate AEGIS on V3C. Grounded VQA accuracy, ambiguity shifts, and visual consistency are assessed via an automated LLM-as-a-judge protocol powered by Gemini 3.7 Flash (High) ($T = 0$), yielding deterministic scoring. In core competition tasks, our full engine achieves an Overall RAG Score of **94.7/100.0** with **88.9%** Recall@1 (100.0% Recall@5, MRR 0.944). Across a 200-query corpus stress test, our full method maintains an Overall RAG Score of **76.2/100.0**, with **57.9%** Recall@1 (74.7% Recall@5, MRR 0.665), **96.7%** Grounded VQA Exact Match, and **100.0%** Fail-Closed Safety.

Table 1. Comprehensive ablation suite across core AEGIS dimensions (Ablations 1–5).

Panel A: Multimodal Retrieval Fusion (200-Query Corpus Test)				
(M1) Dense Only (WeMM-4B)	R@1: 26.3%	R@5: 48.4%	0.354	0.038s
(M2) Dense + Sparse BM25 Payload	R@1: 32.6%	R@5: 53.7%	0.412	0.052s
(M3) Dense + BM25 + SigLIP Secondary	R@1: 38.9%	R@5: 58.9%	0.468	0.076s
(M4) + 4-Way Weighted RRF Fusion	R@1: 44.2%	R@5: 64.7%	0.526	0.084s
(M5) + Temporal Coherence & Diversify	R@1: 51.6%	R@5: 70.5%	0.598	0.091s
(M6) + Parallel VLM Rerank (Full Engine)	**R@1: 57.9%**	**R@5: 74.7%**	**0.665**	1.480s
Panel B: Conversational KIS-C Multi-Turn Dynamics				
(C1) Turn 1: Initial Vague Query	R@1: 0.0%	R@3: 35.0%	0.245	0.088s (Amb 0.84)
(C2) Turn 2: Naive History Concat	R@1: 25.0%	R@3: 50.0%	0.395	0.092s (Amb 0.74)
(C3) Turn 2: + Entity-Preserving CQR	R@1: 50.0%	R@3: 70.0%	0.612	0.110s (Amb 0.59)
(C4) Turn 2: + Compound N-gram Boost	R@1: 75.0%	R@3: 90.0%	0.825	0.115s (Amb 0.42)
(C5) Turn 3: + Negative Filter & Rocchio	**R@1: 85.0%**	**R@3: 95.0%**	**0.910**	**0.120s (Amb 0.24)**
Panel C: VQA Grounding & Fail-Closed Safety				
Ungrounded Whole-Frame VLM	Exact Match: 52.0%	Faithfulness: 58.0%	42.0% Halluc.	1.42s (Hazardous)
Locate-and-Crop (YOLOE-26 + VLM)	Exact Match: 83.3%	Faithfulness: 88.0%	12.0% Halluc.	1.55s
AEGIS Fail-Closed Grounded Contract	**Exact Match: 96.7%**	**Faithfulness: 96.7%**	**0.0% Halluc.**	1.62s (100% Safe)
Panel D: Multi-threaded VLM Concurrency Scaling (Top-10 Scoring)				
Sequential Execution ($N = 1$ worker)	Throughput: 0.67 QPS	14.85s (1.0×)	Instant Screen	—
Parallel Execution ($N = 4$ workers)	Throughput: 2.51 QPS	3.98s (3.73×)	Balanced Live	—
Parallel Execution ($N = 8$ workers)	**Throughput: 5.41 QPS**	**1.85s (8.03×)**	High Ambiguity	—
Panel E: Budgeted Precision Ladder (HNSW Effort Scaling)				
Fast Mode (HNSW $ef = 64$)	Recall: 97.8%	12.4ms	—	Instant Screen
Standard Mode (HNSW $ef = 256$)	Recall: 99.4%	24.5ms	—	Balanced Live
Deep Mode (HNSW $ef = 512$)	Recall: 99.9%	48.6ms	—	High Ambiguity
Exact Brute-Force Scan	**Recall: 100.0%**	118.5ms	—	Deterministic

5.2 Empirical Findings

As documented in Table 1, controlled ablations validate our system: (1) **Fusion Lift**: M1 → M6 raises Recall@1 from 26.3% to 57.9% (+31.6%) and MRR from 0.354 to 0.665; (2) **Conversational Precision**: CQR with compound n-gram boosting reduces ambiguity (0.84 → 0.24) and lifts targets to 85.0% Recall@1 (MRR 0.910); (3) **Zero Hallucination**: fail-closed verification attains 96.7% exact match with 0.0% hallucination; (4) **Sub-2s Latency**: multi-threading yields an 8.03× speedup (1.85s); and (5) **Precision Escalation**: the ladder scales from 12.4ms (97.8% recall) to 118.5ms (100.0% recall).

6 Conclusion

AEGIS introduces an evidence-carrying, live-first multimodal retrieval architecture for VBS 2027. By coupling 4B-parameter WeMM-Embedding representations with Qdrant HNSW indexing, multi-turn compound n-gram clarification boosting, fail-closed grounded VQA, and interactive timeline exploration, the system delivers sub-second responsiveness with verifiable accuracy. Future work targets on-device lightweight VLM distillation and live rehearsal at MMM 2027.

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